

# Chapter 9

## The Obidi Field Equations

### Variational Derivation and Dynamical Structure

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#### 9.1 Introduction

The Local Obidi Action established in the preceding chapter constitutes the dynamical foundation of the Theory of Entropicity. The next step is to derive the equations governing the evolution of geometry and entropy within the Hybrid Manifold Architecture System.

In modern physics, field equations emerge from variational principles. Maxwell's equations emerge from variation of the electromagnetic action. Einstein's equations emerge from variation of the Einstein-Hilbert action. Likewise, the Obidi Field Equations emerge from variation of the Local Obidi Action.

These equations represent the master dynamical laws of the Theory of Entropicity.

They determine:

- the evolution of spacetime geometry,
  - the evolution of informational geometry,
  - the evolution of the entropy field,
  - the coupling between entropy and curvature,
  - the emergence of Einstein gravity.
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#### 9.2 The Fundamental Variational Principle

The Local Obidi Action is

$$I_{LOA} = \sum_J I_{LOA}^{(J)}.$$

For each sector,

$$I_{LOA}^{(J)} = \int \left( R^{(J)} - \Lambda_{(J)} + \kappa_{(J)} g^{(J)\mu\nu} \nabla_\mu^{(J)} S \nabla_\nu^{(J)} S + V(S) \right) e^{S/k_B} \sqrt{|g^{(J)}|} d^4x.$$

The fundamental dynamical requirement is

$$\delta I_{LOA} = 0.$$

This variation must be performed with respect to:

$$g_{\mu\nu}^{(J)},$$

and

$$S.$$

The resulting equations define the dynamics of the theory.

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## Part I

### Metric Variation

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#### 9.3 Variation of the Curvature Term

The variation of the Ricci scalar obeys

$$\delta \left( R \sqrt{|g|} \right) = \left( R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) \delta g^{\mu\nu} \sqrt{|g|} + \text{boundary terms}.$$

Discarding boundary contributions yields

$$\delta \left( R \sqrt{|g|} \right) = G_{\mu\nu} \delta g^{\mu\nu} \sqrt{|g|}.$$

For each sector,

$$\delta \left( R^{(J)} \sqrt{|g^{(J)}|} \right) = G_{\mu\nu}^{(J)} \delta g^{(J)\mu\nu} \sqrt{|g^{(J)}|}.$$

Thus the Einstein tensor naturally emerges from the variation.

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#### 9.4 Variation of the Cosmological Term

The cosmological contribution is

$$-\Lambda_{(J)} \sqrt{|g^{(J)}|}.$$

Using

$$\delta\sqrt{|g|} = -\frac{1}{2}\sqrt{|g|}g_{\mu\nu}\delta g^{\mu\nu},$$

one obtains

$$\delta\left(-\Lambda_{(J)}\sqrt{|g^{(J)}|}\right) = \frac{1}{2}\Lambda_{(J)}g_{\mu\nu}^{(J)}\delta g^{(J)\mu\nu}\sqrt{|g^{(J)}|}.$$

This term contributes directly to the field equations.

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## 9.5 Variation of the Entropic Kinetic Term

The kinetic term is

$$\kappa_{(J)}g^{(J)\mu\nu}\nabla_\mu S\nabla_\nu S.$$

Its variation produces

$$\delta(g^{\mu\nu}\nabla_\mu S\nabla_\nu S) = \nabla_\mu S\nabla_\nu S\delta g^{\mu\nu}.$$

The corresponding contribution becomes

$$\kappa_{(J)}\left(\nabla_\mu S\nabla_\nu S - \frac{1}{2}g_{\mu\nu}(\nabla S)^2\right).$$

This term has precisely the form of a scalar-field stress-energy tensor.

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## 9.6 Variation of the Entropic Potential

The potential contribution

$$V(S)$$

produces

$$-\frac{1}{2}V(S)g_{\mu\nu}.$$

This term acts as an effective vacuum energy.

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## 9.7 Variation of the Exponential Entropic Factor

The exponential coupling

$$e^{S/k_B}$$

introduces a distinctive feature absent from General Relativity.

Metric variation yields an additional contribution proportional to

$$e^{S/k_B} G_{\mu\nu}.$$

Consequently the strength of geometric dynamics depends explicitly upon the local entropy field.

This coupling is one of the defining characteristics of the Theory of Entropicity.

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## Part II

# Entropic Stress-Energy Tensor

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## 9.8 Definition

Collecting all entropy-dependent contributions yields the effective entropic stress-energy tensor

$$T_{\mu\nu}^{ent}.$$

Its general structure is

$$T_{\mu\nu}^{ent} = \kappa \left( \nabla_\mu S \nabla_\nu S - \frac{1}{2} g_{\mu\nu} (\nabla S)^2 \right) - \frac{1}{2} V(S) g_{\mu\nu} + T_{\mu\nu}^{(exp)}.$$

Here

$$T_{\mu\nu}^{(exp)}$$

denotes contributions arising from the exponential entropy coupling.

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## 9.9 Physical Interpretation

The entropic stress-energy tensor plays the role ordinarily occupied by matter.

In General Relativity:

$$T_{\mu\nu}$$

represents matter and energy.

In ToE:

$$T_{\mu\nu}^{ent}$$

represents entropy and information.

Thus entropy acquires the capacity to generate curvature.

This constitutes one of the most important conceptual departures from conventional gravitational theory.

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## Part III

# The Obidi Field Equations

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## 9.10 Sectoral Field Equations

Combining all contributions yields

$$G_{\mu\nu}^{(J)} + \Lambda_{(J)} g_{\mu\nu}^{(J)} = 8\pi G_{(J)} T_{\mu\nu}^{ent(J)}.$$

This equation governs each sector independently.

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## 9.11 Obidi Form

Using the hierarchical indexing notation introduced previously,

$$G_{(\mu\nu)J} + \Lambda_{(J)} g_{(\mu\nu)J} = 8\pi G_{(J)} T_{(\mu\nu)J}^{ent}.$$

This expression constitutes the canonical form of the Obidi Field Equations.

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## 9.12 The Master Equation

Applying the Obidi Addition Rule yields

$$\sum_J (G_{(\mu\nu)J} + \Lambda_{(J)} g_{(\mu\nu)J}) = 8\pi \sum_J G_{(J)} T_{(\mu\nu)J}^{ent}.$$

This equation governs the complete Hybrid Manifold Architecture System.

It may be regarded as the master dynamical equation of the Theory of Entropicity.

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## Part IV

## Entropy Variation

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### 9.13 Entropy Equation of Motion

Variation with respect to

$$S$$

produces a second field equation.

The Euler–Lagrange equation gives

$$\frac{\partial \mathcal{L}}{\partial S} - \nabla_\mu \left( \frac{\partial \mathcal{L}}{\partial (\nabla_\mu S)} \right) = 0.$$

After simplification,

$$2\kappa \square S - V'(S) + \frac{1}{k_B} \mathcal{L} = 0.$$

Here

$$\square = g^{\mu\nu} \nabla_\mu \nabla_\nu$$

is the d'Alembertian operator.

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### 9.14 Sectoral Entropy Equation

For each sector,

$$2\kappa_{(J)} \square_{(J)} S - V'(S) + \frac{1}{k_B} \mathcal{L}^{(J)} = 0.$$

The entropy field therefore evolves dynamically within each sector.

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## 9.15 Physical Interpretation

This equation governs:

- entropy propagation,
- entropy accumulation,
- entropy dissipation,
- entropy-induced geometry.

The entropy field becomes a genuine dynamical degree of freedom.

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## Part V

## Conservation Laws

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## 9.16 Sectoral Conservation

Because

$$\nabla^\mu G_{\mu\nu} = 0,$$

the field equations imply

$$\nabla^\mu T_{\mu\nu}^{ent} = 0.$$

Entropy therefore obeys a conservation law analogous to energy-momentum conservation.

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## 9.17 Entropic Continuity Equation

The conservation law may be interpreted as an entropic continuity equation:

$$\nabla_\mu J_S^\mu = 0.$$

where

$$J_S^\mu$$

is the entropy current.

This relation plays an important role in the interpretation of entropy flow throughout HMAS.

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## Part VI

### Recovery of General Relativity

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#### 9.18 Near-Equilibrium Regime

Consider the limit

$$\nabla_\mu S \rightarrow 0.$$

The entropy field becomes spatially uniform.

Consequently,

$$(\nabla S)^2 \rightarrow 0.$$

The entropy stress-energy tensor simplifies dramatically.

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#### 9.19 Decoupling of Informational Sectors

In this regime:

$$FR \rightarrow 0,$$

$$FS \rightarrow 0.$$

The informational sectors cease contributing dynamically.

Only the Lorentzian sector survives.

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#### 9.20 Emergence of Einstein Gravity

The Obidi Field Equation becomes

$$G_{\mu\nu}^{(L)} + \Lambda g_{\mu\nu}^{(L)} = 8\pi G T_{\mu\nu}.$$

Identifying

$$g_{\mu\nu}^{(L)} = g_{\mu\nu},$$



yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

This is precisely Einstein's Field Equation.

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## 9.21 Interpretation

The emergence of Einstein gravity is not imposed externally.

It arises naturally as a limiting solution of the more general entropic framework.

Thus General Relativity becomes an effective macroscopic description of deeper entropic dynamics.

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## 9.22 The Dynamical Meaning of ToE

The Obidi Field Equations establish the central claim of the Theory of Entropicity.

Geometry is not fundamental.

Entropy is not merely descriptive.

Rather,

$$\text{Entropy} \longrightarrow \text{Entropic Stress-Energy} \longrightarrow \text{Geometry} \longrightarrow \text{Gravity}.$$

The field equations provide the mathematical realization of this principle.

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## 9.23 Summary

The Obidi Field Equations constitute the master dynamical laws of the Theory of Entropicity.

They emerge from variation of the Local Obidi Action and govern the interaction between entropy and geometry throughout the Hybrid Manifold Architecture System.

Their principal features are:

1. Sectoral Einstein tensors.
2. Sectoral cosmological constants.
3. Entropic stress-energy tensors.
4. Dynamical entropy equations.

5. Conservation laws.

6. Recovery of General Relativity in the low-entropic-gradient limit.

These equations transform the conceptual foundations established in previous chapters into a predictive dynamical theory.

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## 9.24 Transition to Information Gravity

The field equations reveal that informational geometry contributes dynamically to spacetime structure.

This observation leads naturally to one of the central ideas of the Theory of Entropicity:

the concept of **Information Gravity**.

The next chapter develops the transition from information geometry to information gravity and examines how the Fisher–Rao and Fubini–Study sectors contribute to emergent gravitational phenomena.